

Infinitesimal Calculus 2, Quiz 3 Simulation

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Do not write at the back of the papers!!

Solution

Lecturer: Dr. Yossi Shamai. Duration: 36 min. **Instructions: Unexplained answers will not receive any credit. You must state all the conditions for every theorem and/or claim. It is forbidden to rely on material that was not learned in the course. You must write your solutions only in the designated areas. Do not write on the back of the papers!**

1. (20 points) Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive numbers and let $p \in \mathbb{R}$. Consider the sequence defined by

$$b_n = \frac{\ln\left(\frac{1}{a_n}\right)}{\ln(n)}, \quad \forall n \in \mathbb{N}$$

Suppose that $\lim_{n \rightarrow \infty} n^{p-b_n} = 1$. Prove that the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $p > 1$.

Solution:

Note that as

$$b_n = \frac{\ln\left(\frac{1}{a_n}\right)}{\ln(n)}, \quad \forall n \in \mathbb{N}$$

then $b_n \ln(n) = \ln\left(\frac{1}{a_n}\right) = -\ln(a_n)$ and therefore

$$-b_n \ln(n) = \ln(a_n) \Rightarrow \ln\left(n^{-b_n}\right) = \ln(a_n) \Rightarrow a_n = \frac{1}{n^{b_n}}$$

We now have

$$\lim_{n \rightarrow \infty} \frac{a_n}{\left[\frac{1}{n^p}\right]} = \lim_{n \rightarrow \infty} \frac{\left[\frac{1}{n^{b_n}}\right]}{\left[\frac{1}{n^p}\right]} = \lim_{n \rightarrow \infty} n^{p-b_n} = 1$$

Thus, the series converges together with the p -harmonic series, so it converges if and only if $p > 1$.